3BLUE1BROWN · ESSENCE OF LINEAR ALGEBRA

Linear Algebra Complete Notebook

A complete study and revision reference for the Linear Algebra step of the AI Research Roadmap. Every required chapter of the series, rewritten as lessons with worked calculations.

What is inside

13 chapters	Chapters 1–9 and 13–16, in order
Full lessons	Plain-language explanation of every concept
Diagrams	Geometric pictures for the visual ideas
Worked examples	Step-by-step calculations in matrix notation
Tests + answers	3 questions per chapter, easy · normal · hard
Formula boxes	Key rules collected at the end of each chapter
Master reference	All formulas in one table + personal error log

Chapters 10–12 (cross products, Cramer's rule) are skipped — not required for machine learning.

AI Research Roadmap · Phase 1 · Stage 1 · Step 1

Vectors

The building block: everything in linear algebra is made of these.

THE IDEA

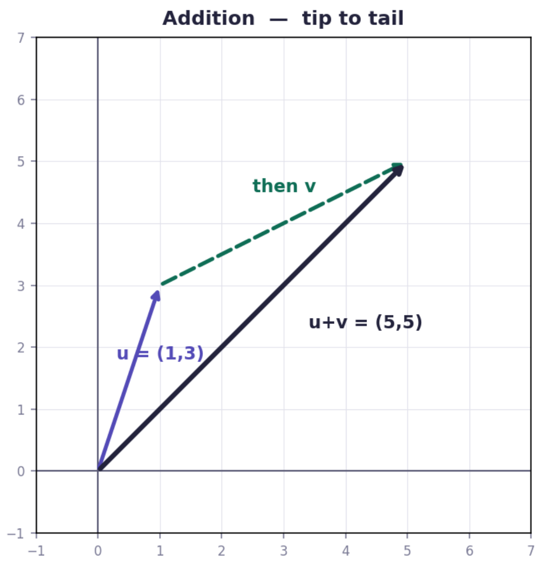
A vector is **not a point** — it is a **movement instruction**. The vector (3, 6) means "go 3 right and 6 up". Because it is only an instruction, you may draw it starting anywhere in the plane and it is still the same vector. Drawing it from the origin is just a convention called **standard position**, useful because then the tip's coordinates equal the vector's numbers.

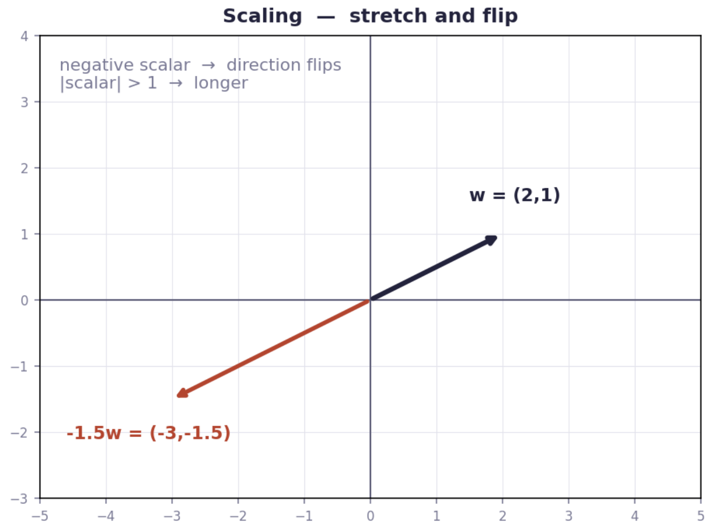
THREE WAYS TO SEE ONE VECTOR

View	What it means
Arrow	A direction and a length drawn on the grid
List of numbers	(3, 6) — easy to compute with
Both	The two views are the same object. Geometry gives meaning, numbers give calculation

THE TWO OPERATIONS

Operation	How	What it does
Addition	Add matching components	Walk the first vector, then the second — tip to tail
Scaling	Multiply every component	Stretch or shrink. Negative scalar flips the direction





TAIL, TIP AND THE THREE FORMULAS

Known	Formula
tail + vector	**tip = tail + vector**
tail + tip	**vector = tip − tail**

tip + vector	**tail = tip − vector**

tail = the start (flat end) · **tip** = the end (arrowhead). Always verify a drawing with tip − tail: it must give back the vector.

UNIT VECTORS AND MAGNITUDE

Magnitude $\|v\|$ is the length: for $v = (3, 4)$, $\|v\| = \sqrt{(3^2 + 4^2)} = \sqrt{25} =$ **5**. A **unit vector** has length exactly 1 — divide each component by the magnitude: $(3/5, 4/5)$. Check it: $(3/5)^2 + (4/5)^2 = 25/25 = 1$ ✓

WORKED EXAMPLES

CHAPTER 1 — WORKED EXAMPLES

ADDITION

$u = (1, 3) \qquad v = (4, 2)$

$$\mathbf{u + v = (1+4,\ 3+2) = (5, 5)}$$

SCALING + SUBTRACTION (write as adding a negative)

$a = (2, -1) \qquad b = (-3, 4)$

$3a = (6, -3)$

$-2b = (6, -8)$

$3a - 2b = (6, -3) + (6, -8)$

$$= (12, -11)$$

NEGATIVE SCALING

$w = (2, 1) \qquad -1.5w = (-3, -1.5)$

direction FLIPS, length × 1.5

RULE: never subtract — turn it into adding a negative first.

> **The habit that prevents most errors:** never subtract directly. Turn every subtraction into *adding a negative* and write that line down before combining.

FORMULAS FROM THIS CHAPTER

Addition	(a,b)+(c,d) = (a+c, b+d)
Scaling	k(a,b) = (ka, kb)
Tip	tip = tail + vector
Vector from 2 points	vector = tip − tail
Magnitude	$\|v\| = \sqrt{(x^2 + y^2)}$
Unit vector	$v / \|v\|$

TEST — Chapter 1

1. u = (1, 3), v = (4, 2). Compute u + v. **→ (5, 5)**
2. a = (2, −1), b = (−3, 4). Compute 3a − 2b. **→ (12, −11)**
3. w = (2, 1). Compute −1.5w, and state the effect. **→ (−3, −1.5); direction flips, length × 1.5**

Linear combinations, span, and basis vectors

What can you build, and what can you reach?

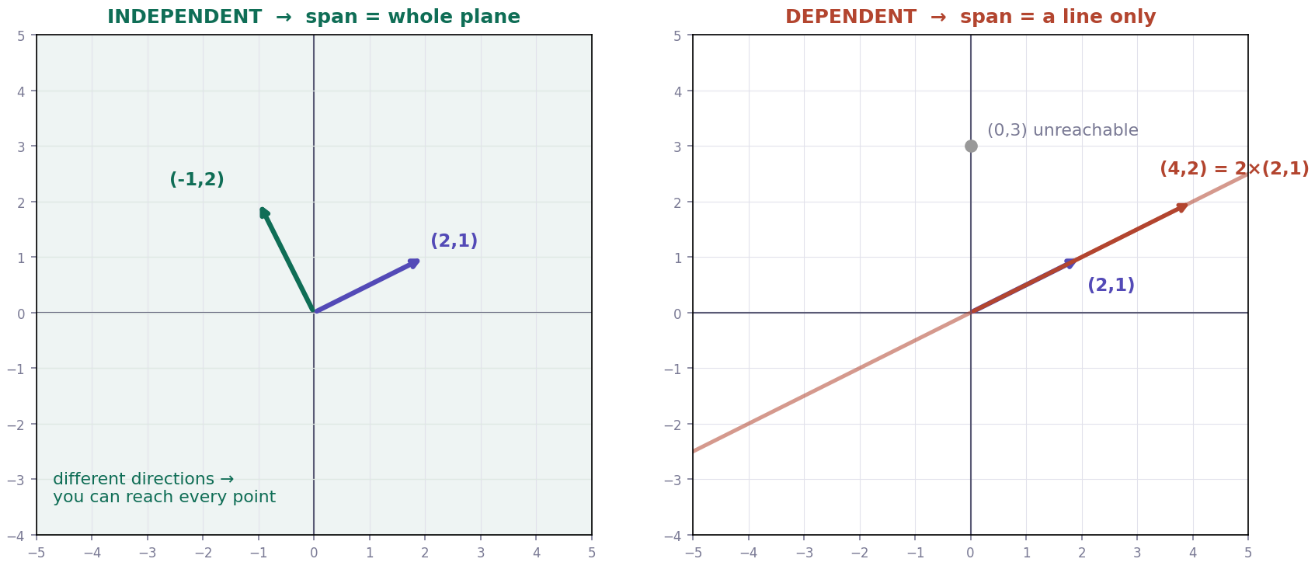
THE IDEA

Take some vectors, **scale each one** by a number, then **add the results**. That is a **linear combination**. The set of *every* point you can reach this way is the **span**.

Term	Meaning
Scalar	An ordinary number that stretches or shrinks a vector
Linear combination	$c_1v_1 + c_2v_2 + \ldots$ — scale each, then add
Span	Every point reachable by linear combinations of the given vectors
Basis	A minimal set of vectors whose span is the whole space (e.g. $\hat{\imath}$ and $\hat{\jmath}$ in 2D)

WHEN SPAN COLLAPSES

Two vectors pointing in **different directions** span the whole plane. But if one is a scaled copy of the other — **parallel** — the span shrinks to a single line, and most of the plane becomes unreachable. That is the situation called **linearly dependent**.



INDEPENDENCE VS DEPENDENCE

Case	Test	Span in 2D
Independent	No vector is a combination of the others	Whole plane
Dependent	One is a scaled copy / combination of the others	A line (or a point)

Quick check in 2D: are they multiples of each other? (2, 1) and (6, 3) are dependent because (6, 3) = 3 × (2, 1).

PARAMETRIC LINES — THE SAME IDEA

A line written $(x, y) = (1, 2) + t(3, -1)$ has two different jobs inside it: **(1, 2) is the anchor** (used once, it fixes where the line sits) and **(3, −1) is the direction** (multiplied by t, repeated at every scale). The dial **t** slides you along the line: $t = 0 \rightarrow (1, 2)$, $t = 2 \rightarrow (7, 0)$.

WORKED EXAMPLES

CHAPTER 2 — WORKED EXAMPLES

LINEAR COMBINATION

$v = (1, 2) \quad w = (3, -1)$

$2v = (2, 4) \qquad 3w = (9, -3)$

$$\mathbf{2v + 3w = (2+9,\ 4+(-3)) = (11, 1)}$$

FINDING UNKNOWN SCALARS

find c1, c2 with $c1(1,2) + c2(2,-1) = (4,3)$

split into components :

$x: \ c1 + 2c2 = 4$

$y: \ 2c1 - c2 = 3$

solve → c1 = 2, c2 = 1

check : $2(1,2) + 1(2,-1) = (2,4)+(2,-1) = (4,3)$ ✓

SPAN TEST

$u = (1,0),\ v = (2,0)$ → both have $y = 0$

any combination = $(c1+2c2,\ 0)$ → y always 0

span = the x-axis only. (0,3) is NOT reachable

> **Why this matters for ML:** if one feature of your dataset is a linear combination of others, it carries no new information — it lies inside the span of what you already have. That is exactly what "redundant" or "collinear" features mean.

FORMULAS FROM THIS CHAPTER

Linear combination	$c_1v_1 + c_2v_2 + \ldots$
Dependent (2D)	one vector is a multiple of the other
Span (dependent)	a line, not the plane
Parametric line	point = anchor + t · direction

TEST — Chapter 2

1. $v = (1, 2)$, $w = (3, -1)$. Compute $2v + 3w$. → **(11, 1)**
2. $a = (2, -4)$, $b = (-5, 1)$. Compute $-3a + 2b$. → **(−16, 14)**
3. Find c_1, c_2 with $c_1(1, 2) + c_2(2, -1) = (4, 3)$. → **$c_1 = 2$, $c_2 = 1$**

Linear transformations and matrices

Where matrices finally make sense: they move space.

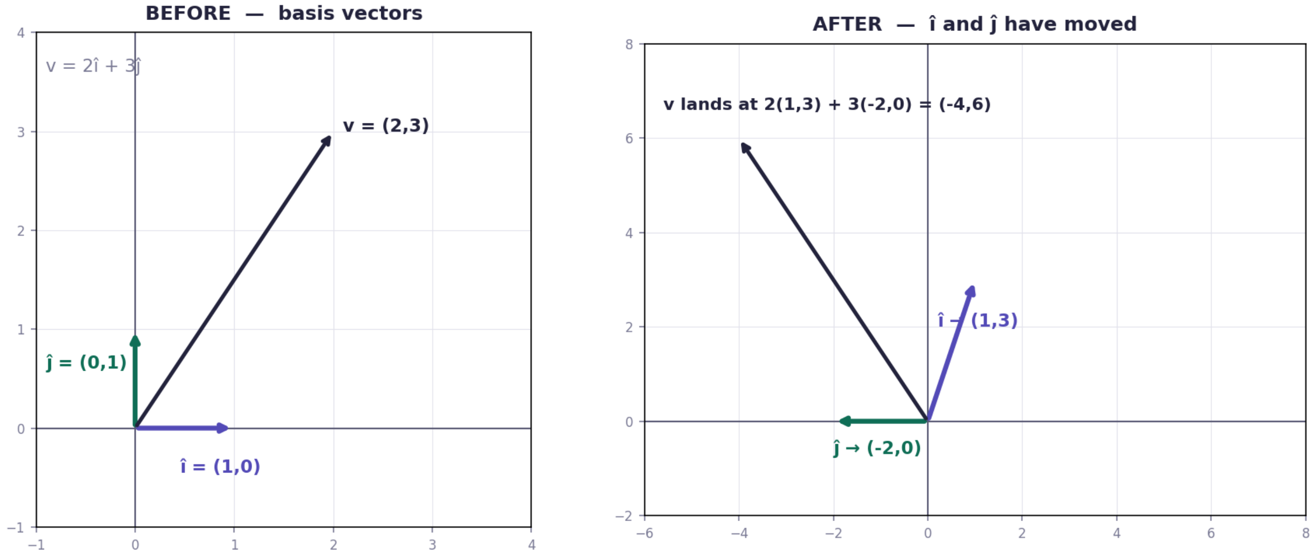
THE IDEA

A **linear transformation** moves every point in space, but keeps grid lines straight, parallel and evenly spaced, with the origin fixed. The remarkable fact: to know where *every* vector goes, you only need to know where **î and ĵ** go.

> **The whole chapter in one line:** the columns of a matrix are where î and ĵ land.

READING A MATRIX

Matrix part	Meaning
Column 1	Where î = (1, 0) lands
Column 2	Where ĵ = (0, 1) lands
Applying it	input (x, y) → x·(column 1) + y·(column 2), then add



WHY IT WORKS

The input (2, 3) literally means "2 steps along î, 3 steps along ĵ". After the transformation, î and ĵ have moved — so the answer is 2 of *where î went* plus 3 of *where ĵ went*. Nothing more.

WORKED EXAMPLE

CHAPTER 3 — WORKED EXAMPLES

matrix columns = where $\hat{\imath}$ and $\hat{\jmath}$ land

RULE

input (x, y) → x·(column 1) + y·(column 2)

EXAMPLE

$\hat{\imath} \to (1, 3)$ $\hat{\jmath} \to (-2, 0)$ input $(-2, 4)$

$$\begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix} \quad \mathbf{x = -2, \quad y = 4}$$

$-2 \times (1, 3) = (-2, -6)$

$4 \times (-2, 0) = (-8, 0)$

add : $(-2 + (-8), \ -6 + 0) = (-10, -6)$

This is matrix × vector — the core operation of every neural network.

FORMULAS FROM THIS CHAPTER

Matrix columns	col 1 = where $\hat{\imath}$ lands · col 2 = where $\hat{\jmath}$ lands
Apply to vector	x·(col 1) + y·(col 2)

TEST — Chapter 3

1. $\hat{\imath} \to (2, 0)$, $\hat{\jmath} \to (0, 3)$. Where does (1, 2) land? **→ (2, 6)**
2. $\hat{\imath} \to (1, 3)$, $\hat{\jmath} \to (-2, 0)$. Where does (−2, 4) land? **→ (−10, −6)**
3. $\hat{\imath} \to (1, -2)$, $\hat{\jmath} \to (3, 1)$. Where does (2, −3) land? **→ (−7, −7)**

Matrix multiplication as composition

Two transformations, one after the other.

THE IDEA

Apply one transformation, then another. The combined effect is itself a single transformation, and its matrix is the **product** of the two. That is what matrix multiplication *is* — not an arbitrary rule.

Point	Detail
Order	**AB** means: apply **B first**, then A. Read right to left, like f(g(x))
Method	Push î through both → column 1. Push ĵ through both → column 2
Not commutative	AB ≠ BA. Rotate-then-shear differs from shear-then-rotate

WORKED EXAMPLE

Matrix multiplication AB

apply B first, then A — read right to left

THE MATRICES

$B = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$ $A = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$ *columns tell you where i and j land*

B sends $\begin{bmatrix} i \end{bmatrix}$ to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} j \end{bmatrix}$ to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

A sends $\begin{bmatrix} i \end{bmatrix}$ to $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} j \end{bmatrix}$ to $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$

STEP 1 — follow i (this gives column 1)

B moves i to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ = 1 step along i, 1 step along j

now apply A to it :

$$1 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1 \times \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

STEP 2 — follow j (this gives column 2)

B moves j to $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$ = -1 step along i, 0 steps along j

now apply A to it :

$$-1 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 0 \times \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

STEP 3 — the two results become the columns of AB

$$AB = \begin{matrix} 2 & 0 \\ 1 & -1 \end{matrix}$$

col 1 from i
col 2 from j

RULE: push i through both matrices -> column 1. Push j through both -> column 2.

> **Copy the numbers first.** Two matrices means twice as many numbers to mix up — write both matrices' columns on your paper, labelled, before any arithmetic.

FORMULAS FROM THIS CHAPTER

Order	AB = apply B first, then A
Method	push î through both → col 1 · push ĵ through both → col 2
Warning	AB ≠ BA

TEST — Chapter 4

1. B: î→(1,0), ĵ→(0,2). A: î→(3,0), ĵ→(0,1). Columns of AB? **→ col1 = (3, 0), col2 = (0, 2)**
2. B: î→(2,1), ĵ→(0,1). A: î→(1,0), ĵ→(−1,2). Columns of AB? **→ col1 = (1, 2), col2 = (−1, 2)**
3. B: î→(1,−2), ĵ→(3,0). A: î→(0,1), ĵ→(−2,1). Columns of AB? **→ col1 = (4, −1), col2 = (0, 3)**

ESSENCE OF LINEAR ALGEBRA · CHAPTER 5

Three-dimensional linear transformations

The same method with one more component.

THE IDEA

Everything from 2D carries over. There is now a third basis vector $\hat{k}$ = (0, 0, 1) alongside $\hat{\imath}$ and $\hat{\jmath}$, so matrices become **3×3** — three columns, one for where each basis vector lands.

2D	3D
$\hat{\imath}$, $\hat{\jmath}$	$\hat{\imath}$, $\hat{\jmath}$, $\hat{\mathbf{k}}$
2×2 matrix, 2 columns	3×3 matrix, 3 columns
x·(col1) + y·(col2)	x·(col1) + y·(col2) + **z·(col3)**

No new concept — only one extra term to track. Composition works identically: push each basis vector through both matrices.

WORKED EXAMPLE

CHAPTER 5 — WORKED EXAMPLE (3D)

same method, one extra component ($\hat{k}$)

RULE

(x,y,z) → x·(col 1) + y·(col 2) + z·(col 3)

EXAMPLE

$\hat{\imath} \to (2,0,1)$ $\quad \hat{\jmath} \to (0,3,0)$ $\quad \hat{k} \to (-1,1,2)$

input (1, 2, −1)

$$1 \times (2, 0, 1) = (2, 0, 1)$$
$$2 \times (0, 3, 0) = (0, 6, 0)$$
$$-1 \times (-1, 1, 2) = (1, -1, -2)$$

add each component :

x : $2 + 0 + 1 = 3$

y : $0 + 6 + (-1) = 5$

z : $1 + 0 + (-2) = -1$

result = (3, 5, −1)

Why it matters: real machine-learning data lives in hundreds or thousands of dimensions. You cannot picture them, but the arithmetic never changes — just more components.

FORMULAS FROM THIS CHAPTER

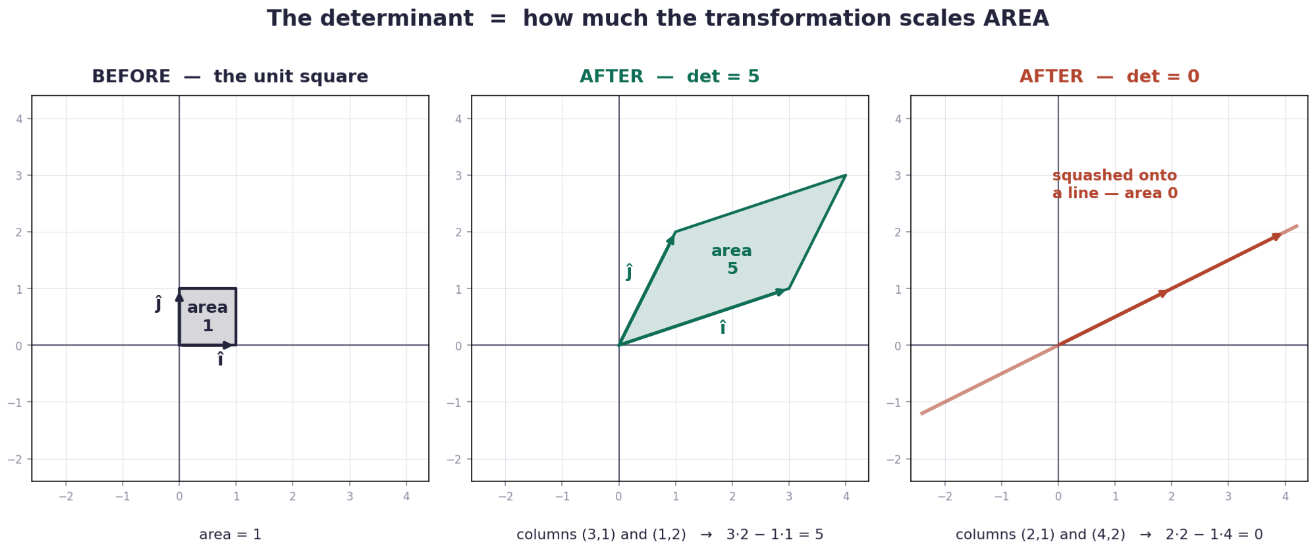
Basis in 3D	$\hat{\imath}, \hat{\jmath}, \hat{k}$ → 3×3 matrix
Apply	$x \cdot (\text{col1}) + y \cdot (\text{col2}) + z \cdot (\text{col3})$

The determinant

One number that says how much space was stretched, squeezed, or destroyed.

THE IDEA

Start with the unit square formed by î and ĵ, area exactly 1. Apply a transformation and it becomes a parallelogram. Its new area is the **determinant** — and every region in the plane is scaled by that same factor, not just that square.



READING THE ANSWER

Determinant	Meaning
det = 3	Areas triple
det = 1	Area unchanged (e.g. a pure rotation)
det = 0.5	Areas halve — space squeezed
det = 0	Space **collapses** onto a line or point — information destroyed
det negative	Space is **flipped over**; the size is |det|

THE FORMULA

For a matrix with columns (a, c) and (b, d):   **det = ad − bc**

WORKED EXAMPLES

CHAPTER 6 — THE DETERMINANT

det = how much the transformation scales AREA

THE FORMULA

for $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$: $\det = a \cdot d - b \cdot c$

a,c = first column b,d = second column

EXAMPLE 1

columns (3, 1) and (1, 2) → $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

$a = 3 \quad b = 1 \quad c = 1 \quad d = 2$

$\det = (3)(2) - (1)(1)$

$= 6 - 1$

$= 5$

areas become 5 times bigger

EXAMPLE 2 (the important case)

columns (2, 1) and (4, 2) → $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$

$a = 2 \quad b = 4 \quad c = 1 \quad d = 2$

$\det = (2)(2) - (4)(1)$

$= 4 - 4$

$= 0$

(4, 2) = 2 × (2, 1) → columns are DEPENDENT

space collapses onto a line → no inverse exists

WHAT THE ANSWER MEANS

det = 3	areas triple
det = 1	area unchanged (e.g. rotation)
det = 0.5	areas halve — squeezed
det = 0	collapsed to a line — NO inverse
det = −2	flipped over, areas doubled

CAREFUL: ad − bc is products THEN subtraction.
Write a, b, c, d labelled first — then compute.

The link that matters: det = 0 means the columns are **linearly dependent** (Chapter 2), space collapses to a lower dimension, two different inputs can land on the same output — so there is **no way back**, and the matrix has **no inverse**. That is the doorway to Chapter 7.

Careful: ad − bc is two products *then* a subtraction. Write a, b, c, d labelled first, then compute.

FORMULAS FROM THIS CHAPTER

Determinant (2×2)	det = ad − bc
Meaning	factor by which areas scale
det = 0	dependent columns · collapse · no inverse
det < 0	space flipped over

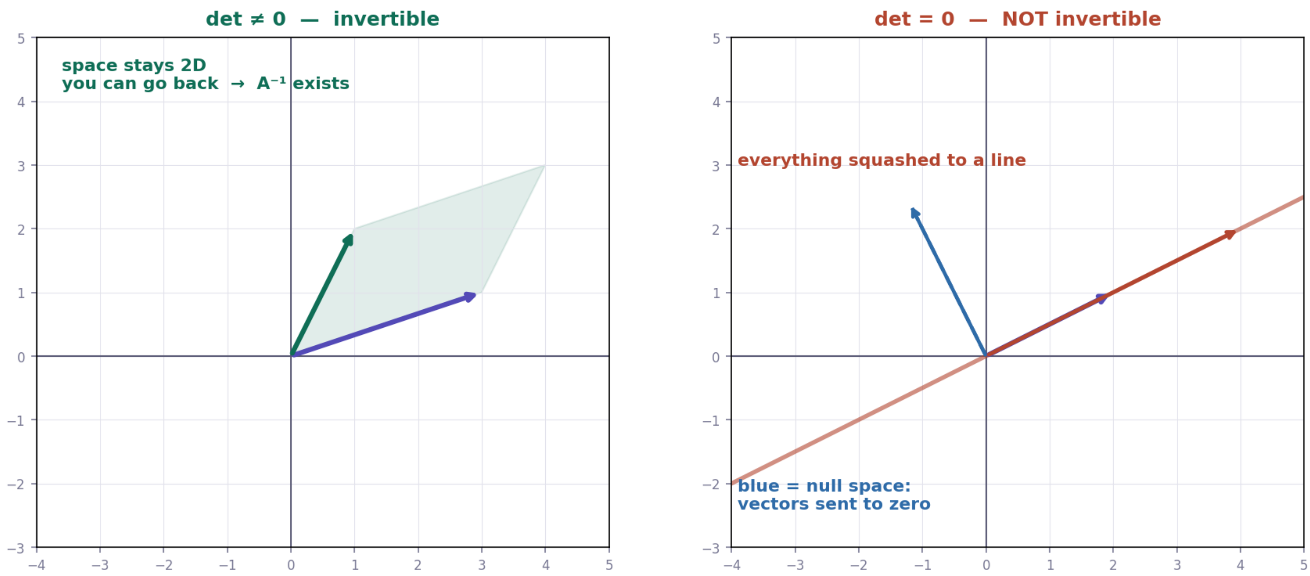
Inverse matrices, column space and null space

Undoing a transformation — and what happens when you can't.

THE IDEA

If a transformation moves space around without collapsing it, there is another transformation that **undoes** it — the **inverse**, written A^{-1}. Applying A then A^{-1} leaves everything exactly where it started. This is how you solve systems of equations: $\mathbf{Ax} = \mathbf{b}$ becomes $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

> **The one condition:** an inverse exists only when **det ≠ 0**. If det = 0, space has been squashed into a lower dimension and information is destroyed — two different inputs land on the same output, so there is no way to reverse it.



THREE WORDS YOU NEED

Term	Meaning
Column space	The span of the matrix's columns — all outputs the matrix can actually produce
Rank	The number of dimensions in that output. Rank 2 in 2D = "full rank"
Null space	All the vectors that get sent to **zero**. Also called the kernel

When det ≠ 0 the null space contains only the origin — nothing is lost. When det = 0 a whole line (or plane) is crushed to zero, and that line *is* the null space.

THE INVERSE FORMULA (2×2)

Swap a and d, flip the signs of b and c, divide everything by the determinant.

WORKED EXAMPLES

CHAPTER 7 — WORKED EXAMPLES

THE INVERSE FORMULA (2×2)

for $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$: $A^{-1} = 1/\det \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

swap a and d, flip the signs of b and c, divide by det

EXAMPLE 1 — find the inverse

$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$

$\det = (3)(2) - (1)(1) = 5$ $\det \neq 0 \rightarrow$ inverse exists

$A^{-1} = 1/5 \times \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$

$= \begin{bmatrix} 2/5 & -1/5 \\ -1/5 & 3/5 \end{bmatrix}$

EXAMPLE 2 — no inverse

$B = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$

$\det = (2)(2) - (4)(1) = 0$

→ columns dependent, space collapses, NO inverse

RANK & NULL SPACE

rank = number of dimensions in the output (column space)

null space = all vectors that get sent to zero

full rank → null space is just the origin → invertible

FORMULAS FROM THIS CHAPTER

Inverse exists	only if $\det \neq 0$
Inverse (2×2)	$A^{-1} = (1/\det) \times [d, -b ; -c, a]$
Solving	$Ax = b \rightarrow x = A^{-1}b$
Rank	dimensions of the output (column space)
Null space	all vectors sent to zero

TEST — Chapter 7

1. A has columns (4, 0) and (0, 2). Find det, then A^{-1}. **→ det = 8; A^{-1} columns (1/4, 0) and (0, 1/2)**
2. A has columns (3, 1) and (2, 1). Find det, then A^{-1}. **→ det = 1; A^{-1} = columns (1, −1) and (−2, 3)**
3. B has columns (6, 3) and (2, 1). Does B^{-1} exist? Justify with a number. **→ det = 6(1) − 2(3) = 0 → no inverse; columns dependent**

Nonsquare matrices as transformations between dimensions

Moving between 2D, 3D and beyond.

THE IDEA

A matrix does not have to be square. A **3×2** matrix takes a 2D vector and produces a 3D vector; a **2×3** matrix does the reverse. The method never changes — each column still says where one basis vector lands.

> **Reading the shape: columns = input dimension**, **rows = output dimension**. A 3×2 matrix has 2 columns (2D input) and 3 rows (3D output).

Shape	Meaning
2×2	2D → 2D, stays in the plane
3×2	2D → 3D, embeds a plane into space
2×3	3D → 2D, projects space down onto a plane
1×2	2D → a single number — this is exactly the dot product

WORKED EXAMPLE

CHAPTER 8 — WORKED EXAMPLES

matrices that move between dimensions

HOW TO READ THE SHAPE

rows = output dimension **columns = input dimension**

a 3×2 matrix : 2 columns → input is 2D, 3 rows → output is 3D

EXAMPLE — 2D input, 3D output

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \\ 0 & -1 \end{bmatrix} \qquad \hat{\imath} \to (2, 1, 0) \qquad \hat{\jmath} \to (0, 3, -1)$$

apply to v = (3, 2) :

$$3 \times (2, 1, 0) = (6, 3, 0)$$

$$2 \times (0, 3, -1) = (0, 6, -2)$$

add : $(6+0,\ 3+6,\ 0+(-2)) = (6, 9, -2)$

a 2D vector became a 3D vector

SHAPES AT A GLANCE

2×2	2D → 2D (same space)
3×2	2D → 3D (into higher dim)
2×3	3D → 2D (projection down)
1×2	2D → 1 number (this is the dot product)

Why it matters for ML: every neural network layer is a nonsquare matrix. A layer mapping 768 features to 256 is a 256×768 matrix — the same operation you just did by hand, at scale.

FORMULAS FROM THIS CHAPTER

Shape	rows = output dim · columns = input dim
3×2	2D → 3D
2×3	3D → 2D
1×2	2D → one number (dot product)

TEST — Chapter 8

1. A 3×2 matrix with $\hat{\imath} \to (1, 0, 2)$, $\hat{\jmath} \to (0, 1, -1)$. Apply it to $(2, 3)$. **→ (2, 3, 1)**
2. A 2×3 matrix with $\hat{\imath} \to (1, 1)$, $\hat{\jmath} \to (0, 2)$, $\hat{k} \to (-1, 0)$. Apply it to $(1, 2, 3)$. **→ (−2, 5)**
3. A layer maps 5 inputs to 3 outputs. What is the matrix shape, and how many numbers does it hold? **→ 3×5, 15 numbers**

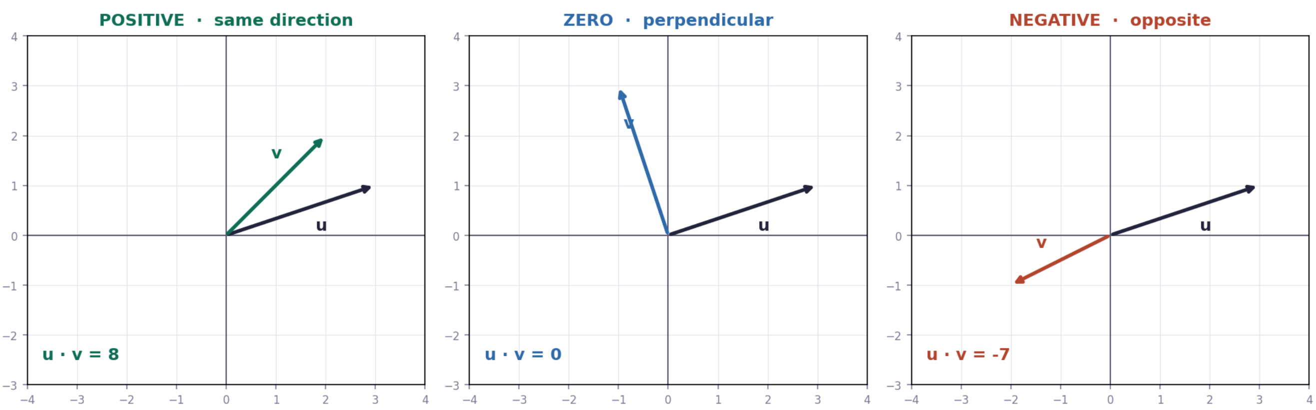
Dot products and duality

Two vectors in, one number out — and what that number tells you.

THE IDEA

The dot product multiplies matching components and adds them, giving a **single number**: (a, b) · (c, d) = ac + bd. Geometrically it measures how much two vectors point *the same way*.

Result	Meaning
positive	The vectors point in the same general direction
zero	They are **perpendicular** (90°)
negative	They point in generally opposite directions



DUALITY — THE SURPRISING PART

Taking the dot product with a fixed vector is the same thing as applying a **1×2 matrix** that squashes 2D down to a number. So every vector has a "twin" that acts as a transformation — that correspondence is what Grant calls **duality**.

WORKED EXAMPLES

CHAPTER 9 — WORKED EXAMPLES

THE FORMULA

$(a, b) \cdot (c, d) = ac + bd$ → the answer is ONE NUMBER

EXAMPLE 1

$u = (3, 1) \quad v = (2, 2)$

$$u \cdot v = (3)(2) + (1)(2) = 6 + 2 = 8$$

positive → they point in the same general direction

EXAMPLE 2 — perpendicular

$u = (3, 1) \quad v = (-1, 3)$

$$u \cdot v = (3)(-1) + (1)(3) = -3 + 3 = 0$$

zero → perpendicular (90°)

SIGN MEANING

> 0	same general direction
= 0	perpendicular
< 0	opposite directions

ML link: cosine similarity = dot product of two UNIT vectors.

> **Why it matters for ML:** this is the most-used operation in machine learning. A neuron computes the dot product of inputs and weights. Similarity between two embeddings is **cosine similarity** — the dot product of the two *unit* vectors (Chapter 1's normalization).

FORMULAS FROM THIS CHAPTER

Dot product	$(a,b)\cdot(c,d) = ac + bd$
Positive	same general direction
Zero	perpendicular
Negative	opposite directions
Cosine similarity	dot product of two unit vectors

TEST — Chapter 9

1. Compute (2, 5) · (3, 1). **→ 11**
2. Compute (4, −2) · (1, 2), and say what the sign means. **→ 0; the vectors are perpendicular**
3. Compute (−3, 1) · (2, −4). **→ −10 (generally opposite directions)**

ESSENCE OF LINEAR ALGEBRA · CHAPTER 13

Change of basis

The same vector, described in someone else's language.

THE IDEA

Coordinates are not absolute — they depend on which basis vectors you use. Another person might use different basis vectors, so the *same arrow* gets different numbers. A **change of basis** translates between the two descriptions.

> **The translation rule.** Let A be the matrix whose columns are *their* basis vectors written in *our* coordinates. Then: **their language → our language: multiply by A. Our language → their language: multiply by A^{-1}.**

TRANSFORMING INSIDE THEIR LANGUAGE

To apply a transformation M while staying in their coordinate system, use **$A^{-1} M A$** — read right to left: translate into our language, transform, translate back. Matrices related this way are called **similar**, and they describe the same underlying transformation seen from two viewpoints.

WORKED EXAMPLE

CHAPTER 13 — WORKED EXAMPLE

same vector, described with different basis vectors

THE TWO DIRECTIONS

their language → our language : multiply by A

our language → their language : multiply by A^{-1}

A = matrix whose columns are THEIR basis vectors (in our coords)

EXAMPLE — their coords → our coords

their basis : $b_1 = (2, 1)$, $b_2 = (-1, 1)$

$A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$ vector in their language : (3, 2)

$3 \times (2, 1) = (6, 3)$

$2 \times (-1, 1) = (-2, 2)$

add : $(6 + (-2),\ 3 + 2) = (4, 5)$

same arrow — just described with our $\hat{\imath}$ and $\hat{\jmath}$

Transformation in their language : $A^{-1} M A$

read right to left : into our language → transform → back to theirs

Why it matters for ML: choosing a good basis is often the whole trick. PCA finds a basis where the data's variance lines up with the axes; embeddings are learned bases where directions carry meaning.

FORMULAS FROM THIS CHAPTER

Their → ours	multiply by A
Ours → theirs	multiply by A^{-1}
A	columns = their basis vectors in our coords
Transform in their language	$A^{-1} M A$

TEST — Chapter 13

1. Their basis: $b_1 = (1, 2)$, $b_2 = (3, 0)$. Their vector (2, 1). What is it in our coordinates? **→ (5, 4)**
2. Their basis: $b_1 = (2, 0)$, $b_2 = (0, 4)$. Their vector (3, 2). Ours? **→ (6, 8)**
3. Their basis: $b_1 = (2, 1)$, $b_2 = (-1, 1)$. Their vector $(1, -2)$. Ours? **→ (4, −1)**

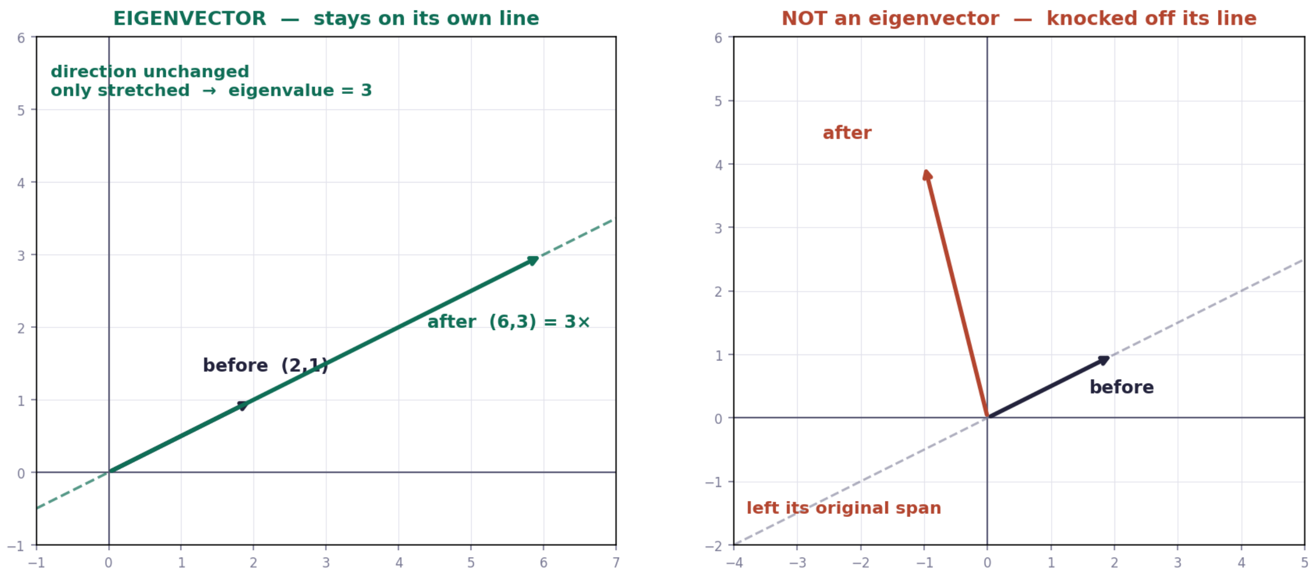
Eigenvectors and eigenvalues

The special directions a transformation cannot knock off course.

THE IDEA

Most vectors get rotated off their original line by a transformation. But some special vectors stay on their own span — they only get **stretched or squashed**. Those are **eigenvectors**, and the stretch factor is the **eigenvalue** λ.

> **The defining equation:** **A v = λ v** — applying the matrix does the same thing as multiplying by a single number.



HOW TO FIND THEM

Step	What to do
1	Write A − λI (subtract λ from each diagonal entry)
2	Set **det(A − λI) = 0** and solve for λ — these are the eigenvalues
3	For each λ, solve (A − λI)v = 0 to get the eigenvector direction
4	Check: A v should equal λ v

Why det = 0? Because A − λI must squash v to zero, and only a collapsing transformation (Chapter 6) can send a non-zero vector to zero.

WORKED EXAMPLE

CHAPTER 14 — WORKED EXAMPLE

find the eigenvalues and eigenvectors of a 2×2 matrix

THE EQUATION

$$A v = \lambda v \qquad \det(A - \lambda I) = 0$$

λ (lambda) = the eigenvalue = the stretch factor

STEP 1 — write A − λI

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \qquad A - \lambda I = \begin{bmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix}$$

STEP 2 — set the determinant to zero

$$\det = (3-\lambda)(2-\lambda) - (1)(0) = 0$$

$$(3-\lambda)(2-\lambda) = 0$$

$$\lambda = 3 \quad \text{or} \quad \lambda = 2$$

these are the eigenvalues

STEP 3 — find the eigenvector for λ = 3

solve $(A - 3I)\, v = 0$:

$$\begin{matrix} 0 & 1 \\ 0 & -1 \end{matrix} \times (x, y) = (0, 0)$$

row 1 : $0x + 1y = 0 \rightarrow y = 0$

x is free → take $x = 1$

eigenvector = (1, 0)

STEP 4 — check it

$$A\,(1, 0) = 1\times(3, 0) + 0\times(1, 2) = (3, 0) = 3 \times (1, 0) \quad ✓$$

ML link : eigenvalues drive PCA, stability of training, and how gradients grow or vanish in deep networks.

Why it matters for ML: eigenvalues are behind PCA (the principal components are eigenvectors of the covariance matrix), the stability of training, and why gradients explode or vanish in deep networks. This is the most ML-relevant chapter in the series.

FORMULAS FROM THIS CHAPTER

Definition	$A v = \lambda v$
Find λ	$\det(A - \lambda I) = 0$
Find v	solve $(A - \lambda I)v = 0$
Check	$A v$ must equal λv

TEST — Chapter 14

1. A has columns (2, 0) and (0, 5). Find both eigenvalues. **→ $\lambda = 2$ and $\lambda = 5$ (diagonal matrix)**
2. A has columns (3, 0) and (1, 2). Find both eigenvalues. **→ $(3-\lambda)(2-\lambda) = 0$ → $\lambda = 3$, $\lambda = 2$**
3. Verify that (1, 0) is an eigenvector of the matrix with columns (4, 0) and (7, 3), and give its eigenvalue. **→ A(1,0) = (4,0) = 4(1,0) → $\lambda = 4$**

A quick trick for computing eigenvalues

A faster route for 2×2 matrices.

THE IDEA

For 2×2 matrices you can skip the full characteristic polynomial. Two facts do the work: the eigenvalues **sum** to a + d (the diagonal, called the trace) and **multiply** to the determinant.

> **The trick.** $m = (a + d)/2$ · $p = ad - bc$ · $\boldsymbol{\lambda = m \pm \sqrt{m^2 - p}}$

Symbol	Meaning
m	Mean of the two eigenvalues = mean of the diagonal
p	Product of the two eigenvalues = the determinant
$\sqrt{m^2 - p}$	How far each eigenvalue sits from the mean

WORKED EXAMPLE

CHAPTER 15 — WORKED EXAMPLE

the fast trick for 2×2 eigenvalues

THE TRICK

$$m = \text{mean of the diagonal} = (a + d) / 2$$

$$p = \text{determinant} = ad - bc$$

$$\lambda = m \pm \sqrt{m^2 - p}$$

EXAMPLE

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$m = (3 + 3) / 2 = 3$$

$$p = (3)(3) - (1)(1) = 8$$

$$\lambda = 3 \pm \sqrt{9 - 8} = 3 \pm 1$$

$$\lambda = 4 \quad \text{and} \quad \lambda = 2$$

CHECK (always do this)

sum of eigenvalues $= 4 + 2 = 6 = a + d$ ✓

product of eigenvalues $= 4 \times 2 = 8 = \det$ ✓

Free self-check: your two eigenvalues must add up to a + d and multiply to the determinant. If either fails, the answer is wrong — no answer key needed.

FORMULAS FROM THIS CHAPTER

m	$(a + d) / 2$
p	$ad - bc$ (the determinant)
Eigenvalues	$\lambda = m \pm \sqrt{m^2 - p}$
Self-check	$\lambda_1+\lambda_2 = a+d \cdot \lambda_1\lambda_2 = \det$

TEST — Chapter 15

1. A has columns (2, 1) and (1, 2). Use the trick. **→ m = 2, p = 3, λ = 2 ± 1 → λ = 3, 1**
2. A has columns (5, 0) and (0, 1). **→ m = 3, p = 5, λ = 3 ± 2 → λ = 5, 1**
3. A has columns (4, 1) and (2, 3). **→ m = 3.5, p = 10, λ = 3.5 ± 1.5 → λ = 5, 2**

Abstract vector spaces

The final step: vectors don't have to be arrows.

THE IDEA

Everything in this series worked with arrows and lists of numbers — but nothing in the reasoning required them. If a set of objects can be **added** and **scaled** sensibly, all of linear algebra applies. Such a set is a **vector space**, and its members are vectors regardless of what they look like.

These are all vectors	Because you can…
Arrows in the plane	add them tip-to-tail, scale their length
Lists of numbers	add componentwise, multiply by a scalar
Functions	add two functions, multiply a function by a number
Polynomials	same — and they have a basis: 1, x, x^2, x^3 …

> **The punchline:** the **derivative is a linear transformation**. Written in the basis 1, x, x^2, x^3, differentiation becomes an ordinary matrix — so calculus can be done with matrix multiplication.

WORKED EXAMPLE

CHAPTER 16 — WORKED EXAMPLE

vectors do not have to be arrows

THE IDEA

anything you can ADD and SCALE sensibly is a vector :

arrows · lists of numbers · functions · polynomials

EXAMPLE — the derivative is a MATRIX

basis of polynomials : $1,\ x,\ x^2,\ x^3$

$p(x) = 3 + 2x + 5x^2 \quad \rightarrow \quad$ coords $(3, 2, 5, 0)$

$$\text{derivative matrix } D = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

apply D to (3, 2, 5, 0) :

row 1 : $0(3) + 1(2) + 0(5) + 0(0) = 2$

row 2 : $0(3) + 0(2) + 2(5) + 0(0) = 10$

rows 3, 4 $= 0$

result (2, 10, 0, 0) → $2 + 10x$

and indeed $\frac{d}{dx}(3 + 2x + 5x^2) = 2 + 10x$ ✓

WHY THE ABSTRACTION PAYS OFF

Because the rules were never about arrows, every theorem you have learned applies anywhere the rules hold — function spaces, probability distributions, quantum states, and the high-dimensional spaces where machine learning lives. That generality is the real reason linear algebra is everywhere.

FORMULAS FROM THIS CHAPTER

Vector space	any set you can add and scale
Polynomial basis	$1, x, x^2, x^3 \ldots$
Key fact	the derivative is a linear transformation

TEST — Chapter 16

1. In the basis 1, x, x^2, x^3, write the coordinates of $p(x) = 4 + 0x + 2x^2$. **→ (4, 0, 2, 0)**
2. Apply the derivative matrix to (4, 0, 2, 0) and give the resulting polynomial. **→ (0, 4, 0, 0) → 4x**
3. Is the set of all 2×2 matrices a vector space? Justify in one line. **→ Yes — you can add two matrices and scale a matrix, so the rules hold**

REFERENCE

Master formula sheet

Ch	Topic	Formula / rule
Ch 1	**Addition**	(a,b)+(c,d) = (a+c, b+d)
Ch 1	**Scaling**	k(a,b) = (ka, kb)
Ch 1	**Tip**	tip = tail + vector
Ch 1	**Vector from 2 points**	vector = tip − tail
Ch 1	**Magnitude**	$\lVert v \rVert = \sqrt{(x^2 + y^2)}$
Ch 1	**Unit vector**	$v / \lVert v \rVert$
Ch 2	**Linear combination**	$c_1v_1 + c_2v_2 + \ldots$
Ch 2	**Dependent (2D)**	one vector is a multiple of the other
Ch 2	**Span (dependent)**	a line, not the plane
Ch 2	**Parametric line**	point = anchor + t · direction
Ch 3	**Matrix columns**	col 1 = where î lands · col 2 = where ĵ lands
Ch 3	**Apply to vector**	x·(col 1) + y·(col 2)
Ch 4	**Order**	AB = apply B first, then A
Ch 4	**Method**	push î through both → col 1 · push ĵ through both → col 2
Ch 4	**Warning**	AB ≠ BA
Ch 5	**Basis in 3D**	î, ĵ, k̂ → 3×3 matrix
Ch 5	**Apply**	x·(col1) + y·(col2) + z·(col3)
Ch 6	**Determinant (2×2)**	det = ad − bc
Ch 6	**Meaning**	factor by which areas scale
Ch 6	**det = 0**	dependent columns · collapse · no inverse
Ch 6	**det < 0**	space flipped over
Ch 7	**Inverse exists**	only if det ≠ 0
Ch 7	**Inverse (2×2)**	A^{-1} = (1/det) × [d, −b ; −c, a]
Ch 7	**Solving**	Ax = b → x = A^{-1}b
Ch 7	**Rank**	dimensions of the output (column space)
Ch 7	**Null space**	all vectors sent to zero
Ch 8	**Shape**	rows = output dim · columns = input dim

Ch 8	**3×2**	2D → 3D
Ch 8	**2×3**	3D → 2D
Ch 8	**1×2**	2D → one number (dot product)
Ch 9	**Dot product**	$(a,b)\cdot(c,d) = ac + bd$
Ch 9	**Positive**	same general direction
Ch 9	**Zero**	perpendicular
Ch 9	**Negative**	opposite directions
Ch 9	**Cosine similarity**	dot product of two unit vectors
Ch 13	**Their → ours**	multiply by A
Ch 13	**Ours → theirs**	multiply by A^{-1}
Ch 13	**A**	columns = their basis vectors in our coords
Ch 13	**Transform in their language**	$A^{-1} M A$
Ch 14	**Definition**	$A v = \lambda v$
Ch 14	**Find λ**	$\det(A - \lambda I) = 0$
Ch 14	**Find v**	solve $(A - \lambda I)v = 0$
Ch 14	**Check**	$A v$ must equal λv
Ch 15	**m**	$(a + d) / 2$
Ch 15	**p**	$ad - bc$ (the determinant)
Ch 15	**Eigenvalues**	$\lambda = m \pm \sqrt{(m^2 - p)}$
Ch 15	**Self-check**	$\lambda_1+\lambda_2 = a+d \cdot \lambda_1\lambda_2 = \det$
Ch 16	**Vector space**	any set you can add and scale
Ch 16	**Polynomial basis**	$1, x, x^2, x^3 \ldots$
Ch 16	**Key fact**	the derivative is a linear transformation

PERSONAL ERROR LOG — WHAT TO WATCH

Error	Fix
Dropping a minus sign	Never subtract — rewrite as adding a negative, on its own line
Misreading given numbers	Copy all given values onto paper, labelled, before computing
Mixing up which column	1st component × 1st column, 2nd × 2nd. Never cross
Rushing the easy question	Same care on all three — misses cluster on the easy one

VERIFY HABITS

- **Vectors:** tip − tail must give back the vector
- **Unit vectors:** components squared must sum to 1
- **Scalars found:** substitute back and check you reach the target
- **Determinant 0:** confirm one column is a multiple of the other
- **Inverse:** $A \times A^{-1}$ must give the identity
- **Eigenvalues:** they must sum to $a + d$ and multiply to det